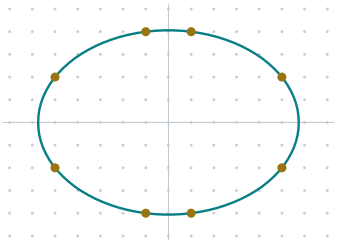
NUMBER THEORY ON ONE PAGE

A dual pair of quadratic fields

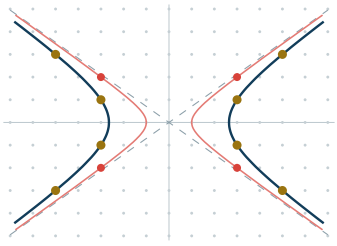
$\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(\sqrt{-2})$, with norms x^2-2y^2 and x^2+2y^2, share exactly the primes that are 1 mod 8.

1 / THE RULE AND THE PICTURE

The norm $N(x+y\sqrt{d})=x^2-dy^2$ is indefinite for $d=2$ and definite for $d=-2$



$x^2+2y^2=33$: 8 points



$x^2-2y^2=7$ gold, $=1$ red

Conjugation sends $\alpha=x+y\sqrt{d}$ to $\bar{\alpha}=x-y\sqrt{d}$, and the norm $N(\alpha)=\alpha\bar{\alpha}$ is multiplicative:

$$\boxed{N(\alpha\beta)=N(\alpha)N(\beta)}$$

For $d=-2$ a level set is an ellipse holding finitely many lattice points and the only units are ±1. For $d=2$ it is a hyperbola holding infinitely many, because the unit $1+\sqrt{2}$ of norm -1 carries points to points, and $3^2-2\cdot2^2=1$ is its square. Both rings $\mathbb{Z}[\sqrt{\pm2}]$ have unique factorisation, discriminants ±8, and 2 ramifies in each.

2 / WHICH PRIMES ARE NORMS

An odd prime is x^2+2y^2 when $p\equiv1,3$ and X^2-2Y^2 when $p\equiv1,7 \pmod 8$

p mod 8	$\mathbb{Q}(\sqrt{-2})$	$\mathbb{Q}(\sqrt{2})$	$\mathbb{Q}(i)$	fields it splits in	a norm
1	yes	yes	yes	all three	$17=3^2+2\cdot2^2=5^2-2\cdot2^2=4^2+1^2$
3	yes	no	no	one only	$11=3^2+2\cdot1^2$
7	no	yes	no	one only	$7=3^2-2\cdot1^2$
5	no	no	yes	one only	$5=2^2+1^2$

The criteria were checked for every odd prime below 3000.
A prime is a norm from a field exactly when it splits there, and the classes 3 and 7 each belong to one field alone, while 1 mod 8 belongs to both.

3 / THE DUALITY

Both fields sit inside $\mathbb{Q}(\zeta_8)$, whose third quadratic subfield is $\mathbb{Q}(i)$

The Galois group of $\mathbb{Q}(\zeta_8)$ is $(\mathbb{Z}/8)^\times=\{1,3,5,7\}$, and each non-trivial element fixes one quadratic subfield: σ_3 fixes $\sqrt{-2}=\zeta+\zeta^3$, σ_7 fixes $\sqrt{2}=\zeta+\zeta^7$ and σ_5 fixes $i=\zeta^2$.
An odd prime p splits in the subfield fixed by σ_a exactly when $p\equiv1$ or $a \pmod 8$, and since $\left(\frac{-2}{p}\right)=\left(\frac{-1}{p}\right)\left(\frac{2}{p}\right)$ each prime splits in exactly one of the three fields, or in all three when $p\equiv1 \pmod 8$.
The real and imaginary fields are therefore partners: the primes 3 mod 8 belong only to $\mathbb{Q}(\sqrt{-2})$, the primes 7 mod 8 only to $\mathbb{Q}(\sqrt{2})$, and $y\mapsto iy$ turns the norm x^2-2y^2 into x^2+2y^2.

4 / ONE PYTHAGOREAN TRIANGLE IN ALL THREE FIELDS

The annulus factors u,v and the hypotenuse c are norms from the three subfields

With $x=m-n$ and $y=n$ in Euclid's formula, the hypotenuse, the inradius and the annulus factors are

$$\boxed{c=(x+y)^2+y^2,\quad u=x^2+2y^2,\quad v=(x+2y)^2-2y^2}$$

the norms of $(x+y)+yi$ in $\mathbb{Q}(i)$, of $x+y\sqrt{-2}$ in $\mathbb{Q}(\sqrt{-2})$ and of $(x+2y)+y\sqrt{2}$ in $\mathbb{Q}(\sqrt{2})$, linked by $u+v=2c$ and $v-u=4xy=4r$.
For $5,12,13$, $(m,n)=(3,2)$ gives $x=1$, $y=2$ and $c=13=3^2+2^2$, $u=9=1^2+2\cdot2^2$, $v=17=5^2-2\cdot2^2$.
Hence the primes 3 mod 8 in $N=uv$ lie in u, those 7 mod 8 in v, those 1 mod 8 may lie in either, and a prime 5 mod 8 can divide only c.

5 / THE SEVEN PRIMES OF $N_0=87\,110\,147\,844\,489$

Four are forced to a side and three are free

prime	mod 8	splits in	norm element and side
3	3	$\mathbb{Q}(\sqrt{-2})$	$1+\sqrt{-2}$, norm $1^2+2\cdot1^2$; goes in u
11	3	$\mathbb{Q}(\sqrt{-2})$	$3+\sqrt{-2}$; goes in u
7	7	$\mathbb{Q}(\sqrt{2})$	$3+\sqrt{2}$, norm $3^2-2\cdot1^2$; goes in v
4583	7	$\mathbb{Q}(\sqrt{2})$	$91+43\sqrt{2}$, norm $91^2-2\cdot43^2$; goes in v
$401,\ 449,\ 457$	1	both	$3^2+2\cdot14^2=23^2-2\cdot8^2$ and similarly for 449, 457; free

Only the three free primes choose a side, giving $2^3=8$ allocations instead of 64 factor pairs.
The two triangles use $u=3\cdot11\cdot401\cdot457$, $v=7\cdot4583\cdot449$ and $u=3\cdot11\cdot449\cdot457$, $v=7\cdot4583\cdot401$.

ASIDE / TRY THESE

$19\equiv3$ is $1^2+2\cdot3^2$; $23\equiv7$ is $5^2-2\cdot1^2$; $13\equiv5$ is neither, but 3^2+2^2 in $\mathbb{Q}(i)$.

LINKS TO OTHER SHEETS

The annulus and conjecture sheets use these rules; the Pell sheet treats the unit $1+\sqrt{2}$.

Convention. $N(\alpha)=\alpha\bar{\alpha}$; $\zeta=e^{i\pi/4}$; the splitting criteria are exact for all odd primes, verified below 3000; (m,n) coprime, $m>n$, opposite parity.

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